

The Arrow's Fingerprint: Event-Density Gravity is Khronometric, GW-Clean, and Lorentz-Safe

Allen Proxmire

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Abstract

Paper GR-I derived Event Density's weak-field Schwarzschild metric and the three classical tests. This paper asks what *class* of gravity ED is — what field equation governs it and how it relates to General Relativity. Three structural results, none requiring the explicit dynamical-bandwidth rule. **(1) The field-equation form is forced.** ED has an emergent metric (Paper GR-I), a covariantly conserved rank-2 source (the bandwidth current's typing is fixed by P05 transport + P04 extensivity, giving $\nabla_\mu T^{\mu\nu} = 0$ as the geodesic condition for matter), and a second-order field law (the Newtonian limit $\nabla^2 b \sim \rho$). By **Lovelock's theorem**, any 4D metric theory with these properties has the field equation $G^{\mu\nu} + \Lambda g^{\mu\nu} = \kappa T^{\mu\nu}$ — *unless* the gravitational field carries a propagating degree of freedom beyond the metric. **(2) It does.** ED's arrow lives in the *law* (P11 irreversibility, P13 tick), not the boundary conditions, so the substrate singles out a preferred foliation at the level of the dynamics. This reduces full diffeomorphism invariance to foliation-preserving diffeomorphisms — the gauge group of **khronometric** (Hořava infrared) gravity — and the mode count is **2 tensor + 1 scalar**: Einstein's tensor sector plus a preferred-foliation “khronon.” ED gravity is khronometric, not pure GR, and the single departure is the fingerprint of the primitive arrow. **(3) The unification protects it.** Because matter and geometry are the *same* bandwidth substrate, every species propagates through one transport (P05) at one cone, so (a) the tensor gravitational-wave speed equals the light speed *structurally* ($c_T = c$, the post-GW170817 constraint as an identity rather than a fine-tuning) and (b) Lorentz violation is **universal, not differential** — removable at leading order, hence safe from the 10^{-20} matter-sector bounds that execute generic Hořava / Einstein-aether theories. The residual, F -dependent quantities — the preferred-frame PPN parameters — are the falsification target (§8). The paper makes no claim of pure GR, no claim that the khronon parameters are computed, and no claim that ED passes the preferred-frame tests.

1. Introduction

1.1 What this paper does

Paper GR-I left the gravitational *class* open: it derived the weak-field metric but not the field equation or the relation to GR. This paper closes that in three steps:

1. **Matter conservation = geodesic motion** (§3), supplying the divergence-free source

Lovelock needs.

2. **The field-equation form via Lovelock** (§4) — forced, conditional on one open condition.
3. **The mode count** (§5) — the open condition fails by one scalar; ED gravity is **khronometric**. Then its two saving graces: **GW-clean** (§6) and **Lorentz-safe** (§7), both structural; and the **residual** falsification target (§8).

1.2 Why this matters

The deepest question about an emergent-gravity program is not “does it recover Newton” (GR-I) but “is it General Relativity, and if not, what is it, and is that survivable.” The honest answer here is the most interesting one: ED is *not* pure GR, it is khronometric, and the deviation is located precisely at ED’s signature commitment — the arrow. That same unification (matter and geometry from one substrate) that makes ED radical is what makes its Lorentz violation safe. The result is a specific, falsifiable theory rather than a vague “GR-like.”

1.3 How this fits the arc

- **Paper GR-I:** weak-field Schwarzschild metric + classical tests (factor of two, redshift, precession). (*inherited*)
- **Paper GR-II (this paper):** field-equation form (Lovelock), gravitational class (mode count $\rightarrow$ khronometric), GW speed, Lorentz safety.
- **Beyond:** the khronon parameters (α_1, α_2) and the MOND-sector relation — model-building, declared open (§8–§9).

1.4 Conventions and regime of validity

Metric signature $(-, +, +, +)$; $c = 1$ where convenient. The structural results — the field-equation form, the mode count, $c_T = c$, universal Lorentz violation — are ***F*-independent**: they follow from the gauge structure and the single substrate, not from the coefficients of the dynamical-bandwidth rule. They are stated in the **continuum (DCGT) limit** and rest on Paper GR-I’s **static, weak-field** background for the explicit checks; the mode count is a linearized statement about perturbations on that background. **No strong-field and no cosmological claim is made** — the cosmological constant Λ is inherited, not a cosmology result — and the *F*-dependent phenomenology $(\alpha_1, \alpha_2, \text{the scalar-mode speed})$ is explicitly deferred (§8).

2. Primitive Inputs

Substrate primitives (postulated, Paper_087): P02 (reciprocal adjacency), P04 (four-band bandwidth, extensive/additive), P05 (adjacency transport), P11 (commitment irreversibility — the arrow), P13 (homogeneous tick — the foliation).

Inherited from Paper GR-I: the emergent metric $g_{ij} \sim b^{-1}$, the lapse $N^2 \sim b$, the weak-field Schwarzschild metric.

Inherited mathematical input:

- **Lovelock’s theorem (1971):** in four dimensions, the only symmetric rank-2 tensor built solely from the metric and its derivatives, of second differential order and linear in the

second derivatives, that is identically divergence-free, is $aG^{\mu\nu} + b g^{\mu\nu}$.

Upstream paper dependencies:

- **Paper_087:** position paper.
- **Paper GR-I:** weak-field metric.
- **Paper_027:** Newtonian limit $\nabla^2 b \sim \rho$; G .
- **Paper_033:** scalar-tensor / MOND arc (sharpened, §9).
- **Paper_073:** DCGT.
- **Paper_089:** V1 finite-width kernel (the single cone / bandwidth-limit speed).

Empirical / value-layer inputs (inherited): $\kappa = 8\pi G$ (G value-inherited, Paper_027); Λ (V1 backreaction). **The dynamical-bandwidth-rule coefficients are not supplied** (the F -residual; §8).

No primitive forcing is invoked.

2.5 Load-Bearing Step Audit

Step	Status	Source / justification
Emergent metric + weak-field Schwarzschild	I	Paper GR-I
Bandwidth current typed as transported-amount (dual) flux	D	§3.1 — forced by P05 transport semantics + P04 extensivity
$\nabla_\mu J^\mu = 0$ (current conservation, covariant)	D	§3.1
$\nabla_\mu T^{\mu\nu} = \rho a^\nu$ (dust); $= 0 \iff$ geodesic motion	D	§3.2 — leading dust model $T^{\mu\nu} = \rho u^\mu u^\nu$
Front = null geodesic (geodesic identity, null sector)	D	§3.3 — discrete Fermat (max- Σ rule)
Lovelock’s theorem	I	standard 4D uniqueness result
ED meets 4D / div-free source / second-order	D / I	§4 — from the rows above + Newtonian $\nabla^2 b \sim \rho$
Field equation forced to $G + \Lambda g = \kappa T$ iff purely metric	D conditional	§4 — Lovelock applied; the “purely metric” condition is §5
Arrow in the law $\rightarrow$ preferred foliation	D	§5.1 — P11/P13 place the time-direction in the dynamics
Gauge = foliation-preserving diffeomorphisms	D	§5.2
Mode count = 2 tensor + 1 scalar (khronometric)	D-structural	§5.3 — gauge-group argument; structural identification, not full linearized propagator (preamble 4)

Step	Status	Source / justification
Single causal cone (one substrate, one transport)	D-structural	§6 — P05 / Paper_089; F -independent
$c_T = c$ (GW170817 passed structurally)	D	§6 — both massless sectors saturate the one cone
Universal (not differential) Lorentz violation	D-structural	§7 — one cone for all species; F -independent
Evades 10^{-20} differential matter-sector bounds	D	§7 — universal LV removable at leading order
$\kappa = 8\pi G$, Λ	I	§4 — value-inherited
Khronon parameters α_1, α_2 ; sub-leading differential LV	deferred (F-dependent)	§8 — model-building; the falsification target

All steps P, D, I, conditional, structural, or deferred. *The two pillars — the field-equation form (Lovelock) and the gravitational class (mode count) — are a standard theorem plus a gauge-group identification; the GW and Lorentz results are F -independent structural consequences of the single substrate; the open quantities are explicitly deferred.*

3. Matter Conservation is Geodesic Motion

Lovelock needs a divergence-free source. ED supplies one, and its conservation turns out to be the geodesic equation.

3.1 The bandwidth current is covariantly conserved — forced

The matter current is the bandwidth adjacency flux J^μ . Its *typing* is fixed, not chosen: by P05 transport semantics it is the *amount transported across an edge* — a flux through the dual face, intrinsically volume-weighted — and by P04 extensivity (bandwidth is an additive capacity) it is a density. A flux of a density has combinatorial (graph) divergence equal to the Levi-Civita covariant divergence $\nabla_\mu J^\mu = (1/\sqrt{g}) \partial_\mu (\sqrt{g} J^\mu)$. So the bandwidth continuity law *is* the covariant conservation

$$\nabla_\mu J^\mu = 0,$$

forced by P05 + P04, with no reference to Einstein. (The naive identification — graph divergence equals covariant divergence — is *false* by a factor $\sqrt{g}$; the correct dual-flux typing repairs it.)

3.2 The rank-2 source and the geodesic condition

The leading rank-2 source from the current is pressureless dust, $T^{\mu\nu} = \rho u^\mu u^\nu$ ($J^\mu = \rho u^\mu$). Its divergence splits:

$$\nabla_\mu (\rho u^\mu u^\nu) = (\nabla_\mu J^\mu) u^\nu + \rho u^\mu \nabla_\mu u^\nu = \rho a^\nu,$$

the first term vanishing by §3.1, leaving ρ times the 4-acceleration a^ν . Therefore

$$\nabla_\mu T^{\mu\nu} = 0 \iff a^\nu = 0 \iff \text{the bandwidth-flux worldlines are geodesics of } g \sim b^{-1}.$$

Matter conservation *is* the geodesic equation — the standard relation, reached forward from ED’s own current. **This equivalence is exact for *dust* ($T^{\mu\nu} = \rho u^\mu u^\nu$); a source with pressure or anisotropic stress adds $\nabla \cdot (\text{stress})$ terms and requires the full dynamical rule’s constitutive content, which is not supplied here.** Dust is the leading, lowest-derivative source the conserved current admits, and it suffices for the Lovelock argument (§4), which needs only *a* divergence-free rank-2 source, not the complete matter sector.

3.3 The geodesic identity holds for the null sector

The substrate front advances to the locally most-coherent / highest-bandwidth channel — a discrete **Fermat principle**, which is null-geodesic propagation. So ED’s fronts *are* null geodesics of the optical metric, and $\nabla_\mu T^{\mu\nu} = 0$ is satisfied for the null sector (the geodesic identity used in Paper GR-I §6 for light bending). The timelike (massive) sector requires the full lapse and is not separately proven here; the dust conservation §3.2 holds formally for any worldlines that are geodesics, which is established for null and assumed for timelike.

4. The Field Equation, via Lovelock

Rather than computing $G^{\mu\nu}$ from b by hand, use the uniqueness theorem that makes Einstein inevitable. **Lovelock’s theorem** (§2): in 4D, the only symmetric rank-2, second-order, identically-divergence-free tensor built solely from the metric is $a G^{\mu\nu} + b g^{\mu\nu}$. So any metric theory meeting those conditions has the field equation $G^{\mu\nu} + \Lambda g^{\mu\nu} = \kappa T^{\mu\nu}$. ED meets three of the four conditions from results already in hand:

Lovelock condition	ED status
4D spacetime	✓ (assembled metric, Paper GR-I)
divergence-free symmetric rank-2 source	✓ $\nabla \cdot T = 0$ (§3); and $\nabla \cdot G = 0$ is the contracted Bianchi identity, automatic for a genuine metric
second-order, linear in second derivatives	✓ at Newtonian order ($\nabla^2 b \sim \rho$); the dynamical-bandwidth rule introduces no higher metric derivatives
built solely from the metric (no extra field)	the open condition — §5

On the **second-order** condition specifically (a reviewer’s natural worry): ED’s dynamical-bandwidth rule $\dot{b} = F(\rho, \nabla\rho, b)$ depends on the source through at most $\nabla\rho$ and on the bandwidth field algebraically and through its Laplacian, so in the continuum (DCGT) limit it introduces **no higher-than-second derivative of the metric** — the Newtonian $\nabla^2 b \sim \rho$ is representative of the full order, not a special low-order case. The Lovelock second-order hypothesis is therefore met at the level the rule is specified.

If the fourth condition holds, the field equation is forced to be Einstein (with Λ). The coefficient $\kappa = 8\pi G$ is fixed by matching the Newtonian limit (G value-inherited, Paper_027); Λ is the V1-backreaction term. This is the legitimate route to $G = \kappa T$: not deriving the tensor, but showing ED satisfies the hypotheses that make it the only possibility — *modulo* the fourth condition, which §5 resolves, and not in ED’s favor for *pure* GR.

Why Lovelock is the right tool. In four dimensions Lovelock’s is the *unique* clean uniqueness theorem for a metric field equation: from general hypotheses (metric-built, second-order, divergence-free) it singles out $G^{\mu\nu} + \Lambda g^{\mu\nu}$ out of the infinite space of candidate tensors. For an *emergent-gravity* program this is exactly what is wanted — it establishes the field-equation *form* from structural facts already in hand (a metric, a conserved source, second-order dynamics) instead of computing $G^{\mu\nu}$ from b by brute force, and it concentrates the entire remaining question into a single hinge: the “purely metric” condition (§5).

5. The Mode Count: ED Gravity is Khronometric

The propagating degrees of freedom of a gravity theory are fixed by its gauge symmetry, and that is where ED departs from pure GR.

5.1 The arrow is in the law, not the boundary conditions

General Relativity’s field equations are time-symmetric and fully diffeomorphism-invariant; a solution acquires an arrow only from its initial/thermodynamic conditions. **ED’s arrow is a primitive of the dynamics** — P11 (irreversible commitment) and P13 (the homogeneous tick) make the time-direction part of the law. The substrate singles out a **preferred foliation** (the slicing by ticks, oriented by the commitment arrow) at the level of the rules.

5.2 The gauge group is foliation-preserving

A preferred foliation in the law breaks full 4D diffeomorphism invariance down to **foliation-preserving diffeomorphisms** (spatial diffeomorphisms on each slice plus time reparametrization). This is exactly the gauge group of **khronometric gravity** — the hypersurface-orthogonal Einstein-aether theory, equivalently the infrared limit of Hořava gravity — in which a scalar field (the “khronon,” whose gradient defines the preferred time) supplements the metric.

Explicitly: a preferred foliation is equivalent to a hypersurface-orthogonal unit timelike vector field $u^\mu = \partial^\mu T / |\partial T|$, where the scalar T (the khronon) labels the leaves; the gauge group reduces from the full $\text{Diff}(M)$ to the foliation-preserving $\text{Diff}(\Sigma) \ltimes \mathbb{R}$ — spatial diffeomorphisms on each leaf Σ plus time reparametrization. This is the standard khronometric reduction, and it is the precise content of “a preferred time direction is a scalar whose level sets are the foliation”: **in ED that scalar is the commitment arrow, so the khronon is not an added field but the arrow made geometric.** Unlike relativistic-MOND constructions such as TeVeS, ED adds **no vector field** — the preferred frame is not an independent dynamical aether but the substrate’s own arrow, which is exactly why (§6–§7) it shares the matter cone instead of carrying separate kinetics.

5.3 The count: 2 tensor + 1 scalar

With foliation-preserving rather than full diffeomorphism gauge, the standard counting gives **2 transverse-traceless tensor modes + 1 scalar mode**. The bandwidth field b_{ij} is a symmetric rank-2 tensor (P02 reciprocity), with the same component count as the metric, so the count is decided by gauge, not by components — and the reduced gauge leaves the extra scalar propagating. That scalar is the **khronon**: in ED it is the substrate’s preferred foliation made dynamical, the arrow promoted to a field.

So **the fourth Lovelock condition fails** — ED’s gravity is *not* built solely from the metric; it carries the khronon. Lovelock therefore does *not* give pure Einstein; it places ED in the **khronometric** class: Einstein’s tensor sector plus the preferred-foliation scalar. ED gravity is khronometric, and the single departure from pure General Relativity is the inevitable shadow of the primitive arrow.

This extra scalar is a concrete, falsifiable prediction, stated here at the point it arises: khronometric gravity carries a **scalar gravitational-wave polarization** — the khronon “breathing” mode — that pure GR forbids, and ED inherits it (§10; falsifier §11.5).

5.4 Consistency with Paper GR-I

Khronometric gravity *shares* GR’s static, spherically-symmetric weak-field metric, so Paper GR-I’s results (factor-of-two bending, redshift, precession; the Schwarzschild relation) are unchanged — ED is observationally GR-like everywhere those tests reach. The departure shows up in the preferred-frame sector, treated below.

6. GW-Clean: the Single Causal Cone

In generic Einstein-aether / Hořava gravity the aether is an *independent* field with its own kinetic term, so its modes have their own light cone, and the post-GW170817 constraint $c_T = c$ (tensor gravitational-wave speed = light speed to 10^{-15}) is a fine-tuning. **ED is different by construction**. It has one maximal signal speed — the bandwidth-limit speed (V1 / Paper_089) — and one transport mechanism (P05 adjacency) that bounds every substrate process. Light is a matter front through P05; a tensor gravitational wave is a perturbation δb_{ij} through P05; there is no separate aether kinetics. Both massless sectors saturate the **same** cone:

$$c_T = c_{\text{light}} \quad \text{structurally.}$$

The constraint that fine-tunes generic Hořava/Einstein-aether is, for ED, an *identity* — its khronon is the substrate’s own foliation, sharing the matter cone. This is F -independent.

One caveat, to avoid overclaiming: the argument above is for the massless **tensor** sector. The **scalar** (khronon) mode may, in principle, propagate at a different speed; whether ED’s single-transport structure forces it onto the same cone as well is an internal-consistency question — a potential strength (maximal predictivity) or an over-constraint against khronometric stability — deferred to §8. The $c_T = c$ result does not depend on how that resolves.

7. Lorentz-Safe: Universal, not Differential, Violation

The constraint that *kills* most Lorentz-violating gravity is not the preferred-frame PPN parameters; it is **matter-sector differential Lorentz violation** — different species having different cones (photon vs electron, vacuum Čerenkov, clock comparison) — bounded at 10^{-20} , into which gravitational-sector violation percolates through loops at order one unless protected. ED’s defining feature turns this aside, *F*-independently:

ED has only one cone, shared by all species (light, electrons, gravitational waves, the khronon all propagate through the same P05 substrate, §6). So ED’s Lorentz violation is **universal** — one deformed cone for everything — and universal violation is removable by a single coordinate/field rescaling, hence **unobservable at leading order**. The 10^{-20} bounds constrain *differential* violation, which ED does not produce.

Put dispersion-theoretically, so the distinction is not hand-waved: **universal** Lorentz violation is a *single* modified dispersion relation shared by all species, removable by one global rescaling and hence unobservable at leading order; **differential** violation is *species-dependent* dispersion — different cones for photon, electron, ... — which is precisely what the 10^{-20} bounds constrain. ED forbids the differential case structurally, because every species *is* the same P05 substrate process and so inherits the same cone.

ED’s unification is its shield. Generic Hořava keeps the aether as a *separate* field, so matter and gravity can drift to different cones (differential violation, executed). ED’s matter and geometry descend from the *same* bandwidth substrate, so there is one cone (universal violation, safe). The very feature that makes ED radical — geometry and matter from one substrate — is what makes its Lorentz violation survivable. This is the strongest viability result of the paper, because it disarms the tightest constraint.

Historical context. Lorentz-violating gravity has a graveyard. Einstein-aether (Jacobson–Mattingly, 2001) and Hořava gravity (2009) introduced a preferred frame for UV or phenomenological gains; most variants were then constrained to death by two facts — **GW170817** (2017) fixed the tensor gravitational-wave speed to the light speed to 10^{-15} , and matter-sector Lorentz tests bound *differential* violation to 10^{-20} . Generic theories survive only by fine-tuning both. ED meets the first as an identity (§6) and the second structurally (§7), for one reason: its preferred frame is the substrate’s own arrow, not a separate aether with independent kinetics. ED is, in effect, the *maximally constrained* khronometric theory — which is why it survives where added-field versions do not.

8. The Residual and the Falsification Target

What survives as open is *F*-dependent and weaker, but real:

- **Preferred-frame PPN parameters** α_1, α_2 (pure-gravity preferred-frame effects; bounds $10^{-4}/10^{-7}$). These are set by the khronon–gravity coupling in the *linearized dynamical-bandwidth rule*, whose coefficients are not pinned here. ED’s preferred frame is naturally the cosmological (CMB) rest frame — which fixes *which* frame the bounds apply in — but does **not** by itself make the couplings small. **This is ED-gravity’s genuine falsification target.**

- **Sub-leading differential Lorentz violation.** Universal violation is leading-order; species couple slightly differently beyond leading order, generating suppressed differential effects whose size is the F -dependent naturalness question.
- **A possible over-constraint.** If every substrate mode shares the one cone (§6), the scalar khronon mode may be forced to c as well; khronometric stability places specific demands on the scalar speed, so this is an internal-consistency check — a potential strength (maximal predictivity) or problem (over-rigidity).

Computing α_1, α_2 requires choosing the dynamical-bandwidth coefficients — model-building, not derivation. This paper deliberately does not do so, and does **not** claim ED passes the preferred-frame tests. The boundary between what ED’s structure forces (the field-equation form, the class, $c_T = c$, universal LV) and what requires the rule’s coefficients (α_1, α_2) is the boundary between this paper and the phenomenology that follows it.

*(Forward note: the dynamical rule and these coefficients are built and analyzed in GR-III/GR-IV, and the front resolves favorably. $\alpha_2 = 0$ **exactly** — a structural consequence of both gravitational cones being luminal ($c_T = c$, $c_s = c$), independent of any tuning and literature-verified, so the tighter bound is met for free — and $\alpha_1 = -4\lambda_{\text{local}}$ is density-suppressed to $\lesssim 4 \times 10^{-93}$ because commitment is sparse (forced by ED’s quantum sector). The “genuine falsification target” named here is met; this paper’s deferral was by scope, not because the front was unresolvable.)*

9. Position Relative to Paper GR-I and the Published Arc

- **Relative to Paper GR-I:** this paper supplies the *class* GR-I deferred. GR-I’s weak-field metric is the regime where khronometric and pure GR coincide; GR-II locates the departure (the khronon) and shows it is safe.
- **Relative to Paper_033 (scalar-tensor / MOND):** the khronometric structure **revises** Paper_033’s implicit Lorentz-covariance (the metric carries a preferred foliation) and **reframes** its flagged “deep-MOND superluminality” as the expected khronon mode (the superluminal/instantaneous scalar of khronometric theories, the origin of universal horizons) — a defect reframed as a feature. The relation of the khronon to Paper_033’s MOND scalar ($a_0 = cH_0/2\pi$, BTFR) is the natural unification target — one khronometric gravity, GR at solar/BH scale and MOND in the deep galactic field via the khronon, with a_0 cosmological from the cosmic-frame anchoring — and is **open, declared, not claimed**.

10. The Wedge — Where ED Diverges from Pure GR

ED’s empirical separation from pure General Relativity is the khronometric sector:

1. **Preferred-frame effects** (α_1, α_2) — absent in GR, present (size F -dependent) in ED; the falsification target (§8).
2. **A scalar gravitational-wave polarization** (the khronon “breathing” mode) — a concrete observable absent in GR’s two tensor polarizations.
3. **Universal horizons** at $b \rightarrow 0$ — a khronometric structure beyond GR’s Killing horizons; and $b \rightarrow 0$ is exactly ED’s emergent horizon locus (the metric degenerates there), so ED’s horizons are candidate khronometric universal horizons.

These are the genuine wedges, and they are sharp: ED predicts a scalar GW mode and preferred-frame structure that pure GR forbids, while remaining GW-speed-clean and Lorentz-safe (§6–§7).

11. Falsification Criteria

11.1 An extra propagating mode beyond 2 tensor + 1 scalar

Falsifier sentence: *A linearization of the dynamical-bandwidth rule yielding a propagating-mode count other than 2 tensor + 1 scalar — e.g. an additional vector mode — would falsify the §5 khronometric identification and force a different gravitational class.*

11.2 Tensor GW speed $\neq$ light speed

Falsifier sentence: *Empirical demonstration that the tensor gravitational-wave speed differs from the light speed ($c_T \neq c$) beyond observational bounds would falsify the §6 single-cone result and the claim that the two sectors share one substrate transport.*

11.3 Differential matter-sector Lorentz violation at leading order

Falsifier sentence: *Empirical observation of differential Lorentz violation between matter species at leading order (e.g. species-dependent cones above the universal-violation residual) would falsify the §7 universal-violation result and ED’s single-cone structure.*

11.4 Preferred-frame parameters above bound

Falsifier sentence: *Computation of α_1, α_2 from the linearized rule yielding values above the solar-system / pulsar bounds ($\alpha_1 \gtrsim 10^{-4}$, $\alpha_2 \gtrsim 10^{-7}$), with the cosmic-frame anchoring assumed, would falsify the viability of ED-khronometric gravity — the §8 falsification target.*

11.5 No scalar GW polarization

Falsifier sentence: *Definitive absence of any scalar gravitational-wave polarization, at sensitivity excluding the khronon mode, would falsify the §5 khronometric class (which requires a propagating scalar).*

12. Position Statement

Event-Density gravity is **Einstein-class but khronometric**. Given the postulated primitives + the emergent metric (Paper GR-I) + the covariantly-conserved source (§3), Lovelock’s theorem forces the field-equation form to be $G^{\mu\nu} + \Lambda g^{\mu\nu} = \kappa T^{\mu\nu}$ **unless** the gravitational field carries an extra propagating mode (§4). It does: the arrow’s residence in the law (P11/P13) reduces the gauge group to foliation-preserving diffeomorphisms, and the mode count is 2 tensor + 1 scalar (§5) — Einstein’s tensor sector plus a preferred-foliation khronon. ED is therefore khronometric, departing from pure GR only at the primitive arrow. The same unification that produces the khronon protects it: one substrate, one transport, one cone gives $c_T = c$ structurally

(§6) and **universal rather than differential** Lorentz violation (§7), so ED evades the 10^{-20} matter-sector bounds that execute generic Hořava / Einstein-aether theories.

What this paper claims. ED’s gravitational class is khronometric (Einstein tensor sector + khronon), the field-equation form is Lovelock-forced given the metric and conserved source, the tensor GW speed equals the light speed structurally, and the Lorentz violation is universal and hence safe from the tightest bounds — all from ED’s structure, independent of the dynamical-rule coefficients.

What this paper does not claim. ED is not pure General Relativity; the mode count is a structural identification, not a completed linearized-propagator computation; the khronon parameters (α_1, α_2) are not computed and ED is not claimed to pass the preferred-frame tests; κ and Λ are inherited; and the khronon $\leftrightarrow$ MOND-scalar relation is open. The residual is model-building — pin the rule, compute the parameters, confront the preferred-frame bounds — not derivation.

Series context. Paper GR-II of the post-pivot curvature-emergence line, companion to Paper GR-I. The remaining work (khronon parameters; the MOND-sector unification) is phenomenology and is declared open.

Appendix: Cross-References and Glossary

Cross-references

- **Position paper:** Paper_087.
- **Paper GR-I:** weak-field Einstein metric (inherited).
- **Paper_027:** Newtonian limit; G .
- **Paper_033:** scalar-tensor / MOND arc (sharpened, §9).
- **Paper_073:** DCGT.
- **Paper_089:** V1 finite-width kernel (the single cone).

Glossary

- **Khronometric gravity.** The hypersurface-orthogonal Einstein-aether theory (Hořava infrared limit): a metric plus a preferred-foliation scalar (khronon). ED’s gravitational class.
- **Khronon.** The propagating scalar whose gradient defines the preferred time; in ED, the substrate foliation (the arrow) made dynamical.
- **Lovelock’s theorem.** The 4D uniqueness result fixing the divergence-free, second-order, metric-built field-equation tensor to $a G^{\mu\nu} + b g^{\mu\nu}$.
- **Single causal cone.** All ED propagation occurs through one transport (P05) at one maximal speed; hence $c_T = c$ and universal Lorentz violation.
- **Universal vs differential Lorentz violation.** Universal: one cone for all species (removable, safe). Differential: species-dependent cones (constrained to 10^{-20}). ED produces universal.
- **Preferred-frame PPN parameters α_1, α_2 .** Pure-gravity preferred-frame effects; F -dependent; the falsification target.

- **Universal horizon.** In khronometric / Hořava gravity, a surface that traps even the instantaneously-propagating khronon (beyond GR's Killing horizon, which traps only light). In ED it is a candidate identity for the $b \rightarrow 0$ locus where the emergent metric degenerates (§10).
- **Scalar (breathing) GW polarization.** The propagating khronon mode — a gravitational-wave polarization absent in pure GR's two tensor modes; a concrete ED prediction (§5.3, §10).

Reader map and open work

Where to look next. - *The weak-field metric and the three classical tests:* GR-I. - *The MOND sector (a_0 , BTFR):* Paper_033 + §9. - *The khronon parameters and scalar-mode speed (phenomenology):* future work (§8). - *Horizon structure ($b \rightarrow 0$, universal horizons):* ED-10 / V5; §10.

Open work (declared, §8–§9): compute α_1, α_2 and the sub-leading differential Lorentz violation from the linearized rule; compute the scalar-mode speed and settle the §6 over-constraint; unify the khronon with the MOND scalar; extend to the strong-field regime and to cosmology.

End of Paper GR-II.

Post-pivot curvature-emergence line. ED gravity is khronometric — Einstein's tensor sector plus a preferred-foliation khronon (the arrow made dynamical) — GW-clean ($c_T = c$, single cone) and Lorentz-safe (universal, not differential, violation; the unification is the shield). Departs from pure GR only at the arrow; the preferred-frame parameters are the open falsification target. Weak-field metric: GR-I.